

ISM4641 Week03 Assignment

Tuples and Sets (No Functions)

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January 27, 2025

Introduction

Welcome to the **Week 03** assignment in ISM4641! We will focus on **Tuples** (for storing fixed data that shouldn't be changed) and **Sets** (for managing unique collections of items). You can also apply any of the concepts learned so far (loops, conditional statements, etc.)—but **do not** define or use any custom functions yet, since that material will come in Week 4.

The Summit Exports Scenario

Summit Exports is a fictional logistics company shipping specialized items to various destinations. Your tasks:

1. **Use Tuples** to store information about containers (container IDs, coordinates, fragile status, etc.).
2. **Use Sets** to manage the list of unique destinations.
3. **Apply loops** (such as `for` or `while`) and conditional statements (`if`, `elif`, `else`) to demonstrate membership checks and data manipulations in sets.
4. **Present** the results in a clear, step-by-step manner, using print statements and f-strings.

You will write your code in a **Google Colab notebook**, with Markdown cells explaining each step.

Task 1: Container Data with Tuples

1. Create Three Container Tuples

- Each tuple might look like:

```
1 container_a = ("CONT-101", 40.7128, -74.0060, True)
```

Where:

- CONT-101 is the container ID (a string).
- 40.7128 and -74.0060 are the lat/long coordinates (floats).
- True indicates whether the contents are fragile (Boolean).

2. Print Each Tuple

- Show that all three tuples were created properly.

- For instance:

```
1 print("Container A:", container_a)
2 print("Container B:", container_b)
3 print("Container C:", container_c)
```

3. Tuple Immutability Check

- Attempt to modify one of the tuples, for example:

```
1 # container_a[1] = 41.0
```

- Catch or show the error that occurs in a `try-except` block, or comment out the line, and explain in Markdown why it fails.

4. Unpack a Tuple

- Pick one of the container tuples and unpack it into separate variables (e.g. `cont_id, lat, lng, fragile = container_a`).
- Print each variable individually using f-strings to confirm you've successfully extracted the data.

Goal: Demonstrate you understand how to define tuples, that they're immutable, and that you can unpack them into separate variables for clarity.

Task 2: Managing Destinations with Sets

1. Create a Set of Destinations

- For example:

```
1 destinations = {"New York", "Los Angeles", "Dallas", "Chicago", "Miami"}
```

- Print the set. (Remember, sets may appear in any order.)

2. Modify and Check Membership

- **Add** a new city to the set (e.g., "Seattle") using `.add()`.
- **Remove** a city (e.g., "Dallas") using `.remove()`. Print the set after each operation.
- Check membership for a city that might not be in the set (e.g., "Boston") using `"Boston" in destinations`, and print a statement like `Boston is in the set: False/True`.

3. Remove vs. Discard

- Demonstrate the difference between `.remove()` and `.discard()` by calling them on a city that does **not** exist. For example:

```
1 # Attempt remove()
2 destinations.remove("Paris") # Should raise an error if not in the set
3
4 # Attempt discard()
5 destinations.discard("Paris") # Should not raise an error
```

- Print appropriate messages indicating the outcome.

4. Basic Set Operations

- Create two additional sets, for instance:

```
1 west_coast = {"Los Angeles", "Seattle"}
2 east_coast = {"New York", "Miami"}
```

- Print the results of:
 - **Union** of `west_coast` and `east_coast`
 - **Intersection** of `destinations` and `east_coast`
 - **Difference** between `destinations` and `west_coast`

- Symmetric difference between `west_coast` and `east_coast`

Goal: Confirm your ability to manipulate sets, remove duplicates, check membership, and perform set operations.

Task 3: Bringing It All Together

Summit Exports needs a **combined view** of these container tuples and the shipping destinations.

1. Display Container Details

- Print each container's info in a user-friendly format. For example:

```
Container A:  
  ID: CONT-101  
  Coordinates: (40.7128, -74.0060)  
  Fragile: True
```

- Use f-strings for clarity.

2. Count Unique Destinations

- Print how many unique destinations are in your `destinations` set using `len(destinations)`.
- To see them in alphabetical order, convert to a list and sort:

```
1 sorted_destinations = sorted(destinations)  
2 print("All destinations (sorted):", sorted_destinations)
```

3. Optional Looping / Condition

- (Optional) Add a small loop that iterates over each destination in `destinations`. Print a statement like:

```
1 for city in destinations:  
2     if city in east_coast:  
3         print(f"{city} is on the East Coast.")  
4     elif city in west_coast:  
5         print(f"{city} is on the West Coast.")  
6     else:  
7         print(f"{city} is neither East nor West Coast, or is unclassified.")
```

Goal: Show that you can combine everything into one scenario, referencing your container tuples and sets of destinations to provide a final summary.

Example (Simplified) Output

Click to expand sample output

--- Task 1: Tuples ---

Container A: ('CONT-101', 40.7128, -74.006, True)

Attempting to modify container_a:

Error: 'tuple' object does not support item assignment

Unpacked Container B ->

 ID: CONT-102

 Lat: 34.0522

 Lng: -118.2437

 Fragile: False

--- Task 2: Sets ---

Initial destinations: {'Chicago', 'Miami', 'Dallas', 'Los Angeles', 'New York'}

After adding Seattle: {'Seattle', 'Miami', 'Dallas', 'Los Angeles', 'New York', 'Chicago'}

Removing 'Dallas'...

destinations = {'Seattle', 'Miami', 'Los Angeles', 'New York', 'Chicago'}

Is 'Boston' in destinations? False

Trying remove('Paris'): KeyError encountered

Trying discard('Paris'): No error

Union West+East: {'Los Angeles', 'Seattle', 'New York', 'Miami'}

Intersection with east_coast: {'New York', 'Miami'}

Difference from west_coast: {'New York', 'Miami', 'Chicago'}

--- Task 3: Combined View ---

Container A:

 ID: CONT-101

 Coordinates: (40.7128, -74.006)

 Fragile: True

...

Number of unique destinations: 5

All destinations (sorted): ['Chicago', 'Los Angeles', 'Miami', 'New York', 'Seattle']

Checking if each city is East or West Coast...

Seattle is on the West Coast.
Miami is on the East Coast.
Los Angeles is on the West Coast.
New York is on the East Coast.
Chicago is neither East nor West Coast, or is unclassified.

Submission

1. Submit on Canvas

- Upload your **Google Colab notebook (.ipynb)** by the deadline.

2. Naming Convention

- `ISM4641_Week03_Assignment_[YourName].ipynb`

3. Code Formatting & Clarity

- Use Markdown headings, bullet points, and code comments to keep your work well-organized.
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Grading Rubric

Criteria	Points
Tuples: creation, immutability demonstration, unpacking	25
Sets: creation, add/remove vs. discard, membership checks, set ops	35
Scenario Completion: clarity of container data + destinations	20
Code Readability: loops/branching used effectively, f-strings, Markdown organization	10
Execution & Testing: runs without error, demonstrates scenario variety	10
Total	100

Note: Provide explanatory comments and/or Markdown cells so it's clear what each block of code does. Good luck exploring **tuples** and **sets**—two essential data structures for business analytics in Python!